Cody Slater

New York, NY codyslater.github.io

About

My aim is closed-loop control of biology: designing each perturbation together with the model that learns from it, so that a cell type can be read, predicted, and steered. I came to it from aerospace, where control of a complex system is the whole problem.

Education

Doctor of Medicine — New York, NY
Columbia University — Expected 2027

Doctor of Philosophy, Biomedical Engineering — New York, NY
Columbia University — Expected 2027

Master of Philosophy, Biomedical Engineering — New York, NY
Columbia University — Expected 2026

Master of Science, Biomedical Engineering — New York, NY
Columbia University — May 2023

Bachelor of Science, Aerospace Engineering — Worcester, MA
Worcester Polytechnic Institute — May 2016

Research & Teaching Experience

Vicković Technology Innovation Lab, New York Genome Center, Graduate Researcher (Ph.D.) — New York, NY
deep learning · PyTorch · GPU HPC · single-cell & spatial transcriptomics · lentivirus · AAV · Cas9 · Cas13 — 2023 – present

- Leading development of large specialized models for RNA modification prediction and generative design.
- Established the first selection-based functional assay for RNA modifications; 10,000+ construct pools.
- Engineering AAV vectors for cell-type-specific perturbation in mouse brain, from prep to tissue processing.
- Building a spatial transcriptomics toolset to model in vivo cell-type-specific genetic perturbations.

Department of Neurological Surgery, Columbia University, Graduate Researcher — New York, NY
intraoperative microneurography · DBS electrodes · human trigeminal ganglion · OR hardware design — 2024 - present

- Built the intraoperative recording setup for human trigeminal ganglion, using DBS electrodes.

Trends in Genomics, Columbia University, Teaching Assistant — New York, NY

- Co-designed the course for its first offering (2025). — Spring 2025 & 2026

Neural Engineering and Control Lab, Columbia University, Graduate Researcher (M.S.) — New York, NY
Neuropixels · GRAB sensors · 2-photon · optogenetics · chemogenetics · stereotaxic surgery · histology · behavior — 2018 – 2023

- Co-created neurOS, a Simulink Real-Time platform for multimodal closed-loop experiment control.
- Designed and implanted brain interfaces for Neuropixels, fiber photometry and 2-photon imaging.
- Developed the lab's in vivo perturbation assay and its cell-type-specific transcriptomic readout.
- Launched a longitudinal locus coeruleus stimulation study in an Alzheimer's model; basis of an R01 line.

Department of Biomedical Informatics, Columbia University, Student Researcher — New York, NY
retrospective case series · clinical data analysis — 2020

- Co-led 30 medical students and 18 clinicians characterizing the first COVID-19 wave in New York.

Professional History

MIT Lincoln Laboratory / MIT Haystack Observatory, Engineer, Advanced Sensors and Techniques Group — Westford, MA
radar systems · RF hardware · thermal design · field deployment · computer vision — 2016 – 2018

- Designed the structure and cooling for a solid-state radar transmitter, fielded in the Marshall Islands.
- Built a stand-alone traveling-wave-tube test facility; led the Millstone 84-foot azimuth encoder upgrade.
- Led a team adapting computer vision for autonomous UAV flight through an obstacle course.

MIT Lincoln Laboratory, Student Researcher, Systems Engineering Group — Lexington, MA
systems design · spacecraft propulsion · vacuum testing — 2015 – 2016

- Designed, built and characterized an experimental pico-satellite monopropellant thruster in vacuum.

GE Aviation / Unison Industries, Aerospace Engineering Intern — Jacksonville, FL
manufacturing engineering · design automation — 2014

- Automated the build instructions for jet-engine cable harnesses, generated from the design files.

Publications & Papers

Intraoperative microneurographic recording from the human trigeminal ganglion using standard deep brain stimulation electrodes
C. Slater, …, R. F. Sekula Jr.
(Submitted (2026))

Intracranial delivery of neuron-preferential lipid nanoparticles for gene-editing activity in mouse brain
S. S. Cai, **C. Slater**†, …, K. W. Leong

10.1016/j.xcrp.2026.103514 (Cell Reports Physical Science 7(9), 103514 (2026))

Behavioral state-dependent norepinephrine dynamics in the primary somatosensory and prefrontal cortices during tactile detection tasks
C. Kelley, T. Lantin, **C. Slater**, …, Q. Wang
10.64898/2026.01.16.699887 (bioRxiv preprint (2026))

Alpha modulation of spiking activity across multiple brain regions in mice performing a tactile selective detection task
C. Kelley, **C. Slater**, …, Q. Wang
10.1111/ejn.70356 (European Journal of Neuroscience 62(12), e70356 (2025))

Bidirectional modulation of somatostatin-expressing interneurons in the basolateral amygdala reduces neuropathic pain perception in mice
A. Apte, J. Fernald, **C. Slater**, …, Q. Wang
10.3389/fpain.2025.1602036 (Frontiers in Pain Research 6, 1602036 (2025))

Brain-wide cell-type-specific noradrenergic modulation of the transcriptome
C. Slater, …, Q. Wang
10.1101/2025.05.09.653136 (bioRxiv preprint (2025))

The Neuromodulatory Role of the Noradrenergic and Cholinergic Systems and Their Interplay in Cognitive Functions: A Focused Review
C. Slater, …, Q. Wang
10.3390/brainsci12070890 (Brain Sciences 12(7), 890 (2022))

Alzheimer's disease: An evolving understanding of noradrenergic involvement and the promising future of electroceutical therapies
C. Slater, Q. Wang
10.1002/ctm2.397 (Clinical and Translational Medicine 11(4), e397 (2021))

Characterising the long-term clinical outcomes of 1190 hospitalised patients with COVID-19 in New York City: a retrospective case series
S. M. Shoucri, et al.
10.1136/bmjopen-2021-049488 (BMJ Open 11(6), e049488 (2021))

Characterization and clinical course of 1000 patients with coronavirus disease 2019 in New York: retrospective case series
M. G. Argenziano, S. L. Bruce, **C. L. Slater**†, …, R. Chen
10.1136/bmj.m1996 (BMJ 369, m1996 (2020))

Interventions to reduce short-wavelength ("blue") light exposure at night and their effects on sleep: A systematic review and meta-analysis
A. Shechter, K. A. Quispe, J. S. Mizhquiri Barbecho, **C. Slater**, L. Falzon
10.1093/sleepadvances/zpaa002 (Sleep Advances 1(1), zpaa002 (2020))

An Interprofessional Approach to Clinical Workflow Evaluation Focused on the Electronic Health Record Using Time Motion Study Methods
J. Schwartz, J. Elias, **C. Slater**, K. Cato, S. C. Rossetti
(Paper presented at the AMIA Annual Symposium Proceedings, 1187–1196 (2019))

Six-Axis Monopropellant Propulsion System for Picosatellites
C. Slater, M. Creaghan, O. Lamce
(Unpublished Major Qualifying Project. Worcester Polytechnic Institute (2016))

Developing a Collaborative Process for the Application of Appropriate Technology on Farms in Mandi District
A. S. Konieczny, **C. Slater**, S. M. Neeno
(Unpublished Interactive Qualifying Project. Worcester Polytechnic Institute (2015))

Fellowships & Grants

Van C. Mow Fellowship — Sept 2021
Established in 2018 by Yuling Wei to provide fellowships to doctoral students enrolled in Columbia's School of Engineering and Applied Science, who are or plan to be engaged in biomedical research.

National Institute on Aging T35AG044303 — Mar 2019
Brief Research in Aging and Interdisciplinary Neurosciences (BRAIN) Program

Presentations

Whole brain, cell type specific transcriptomic profiling of in vivo neurochemical perturbation — Chicago, IL Oct 2024
Society for Neuroscience. Presenting author.

Leveraging large-scale electrophysiology array recording to identify neural biomarkers for improved closed-loop neuromodulation for chronic pain — Chicago, IL Oct 2024

Society for Neuroscience. Contributing author.

Noradrenergic modulation of population activity in the primary somatosensory cortex, striatum, and amygdala in a tactile selective detection task — Chicago, IL, Oct 2024
Society for Neuroscience. Contributing author.

Locus coeruleus modulation of population activity in the awake somatosensory cortex — Washington, DC, Nov 2023
Society for Neuroscience. Contributing author.

Lipid nanoparticle-mediated gene editing in the mouse brain via intracranial delivery — Seattle, WA, Oct 2023
Biomedical engineering society. Contributing author.

Noradrenergic dynamics in the sensory and prefrontal cortices during a perceptual decision making task — San Diego, CA, Nov 2022
Society for Neuroscience. Presenting author.

Functional noradrenergic regulation of the transcriptome during decision making under varying uncertainty — New York, NY, Oct 2022
Center of Excellence in the Neuroscience of Decision-Making Kickoff, Columbia University. Presenting author.

Development of a Neural Network Architecture for Optimal Prediction of EEG Alpha Wave Power — New York, NY, Apr 2022
Northeast Bioengineering Conference, Columbia University. Presenting author.

Targeted Neural Stimulation to Improve Neuropathic Pain Treatment — New York, NY, Apr 2022
Northeast Bioengineering Conference, Columbia University. Presenting author.

MIDA-based Finite Element Modeling to Optimize Electrotherapy Configurations — New York, NY, Apr 2022
Northeast Bioengineering Conference, Columbia University. Contributing author.

Long-term Locus Coeruleus Stimulation reduces Aβ pathology in a Mouse Model of Neurodegeneration — New York, NY, May 2021
Medical School Research Symposium, Columbia University. Awarded best session poster by panel of judges. Presenting author.

Synthetic Neuromodulation as a Potential Treatment for Alzheimer's Disease — New York, NY, Aug 2019
NIH Summer Research Symposium, Columbia University.

The Temporal Expression of Consciousness — Worcester, MA, Mar 2017
Class of 1879 Award Ceremony and Reception, WPI.

The Importance of Biological Processes in Learning — Worcester, MA, Feb 2016
New England Peer Tutor Association Conference.

Analysis and Design of a Solar Powered Cooling System — Lexington, MA, Aug 2015
Technology Office Challenge Poster Presentation, MIT Lincoln Laboratory.

Developing a Collaborative Process for the Application of Appropriate Technology — Mandi, India, May 2015
Interactive Qualifying Project Symposium, Indian Institute of Technology.

Solar Re-Revolution: Engineering the Future in Solar Sustainability — Worcester, MA, Dec 2012
Grand Challenges Seminar Poster Presentation, WPI.

Leadership

Innovative Medicine Club Senior Advisor — Aug 2020 – Aug 2022

President of Columbia Health Tech Assembly — June 2020 – Aug 2022

Medical School Class Council Ethics Representative — Aug 2018 – July 2021

VP&S Curriculum & Education, Discover Subcommittee, Student Member — Jan 2020 – present

President of the WPI Fitness Club — Jan 2015 – Jan 2016

Treasurer WPI Chapter of AIAA — Jan 2014 – Jan 2015

Honors & Awards

Van Mow Fellow — 2021 – 2022

PS250 Roy and Diana Vagelos Scholarship Recipient — 2018 – 2021

D. & D. Merksamer Scholarship Recipient — 2018 – 2021

WPI Class of 1879 Prize for Outstanding Original Work in the Humanities and Arts — Mar 2017

WPI Graduation with Honors and High Distinction — May 2016

WPI Presidential Scholarship Recipient — 2012 – 2016

Mid-Plains Rural Telephone Cooperative Scholarship Recipient — 2012 – 2016

Voted Most Impactful Interactive Socio-Technical Practicum by Panel of Judges — May 2015

WPI Dean's List — Spring 2013 – Fall 2015